

# Ontology-Grounded Semantic Entity Retrieval for Digitized Traditional Vietnamese Medicine Literature: A Chatbot-Integrated Framework

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**Abstract.** Extracting knowledge from Traditional Medicine (TM) remains challenging due to the unstructured nature of historical texts and the complex, dialectical relationships among herbs, formulas, and therapeutic effects. Conventional text-based Retrieval-Augmented Generation (RAG) and keyword search methods often return fragmented results, while generic Large Language Models (LLMs) are prone to generating unverified medical information due to hallucination. To address these limitations, this paper proposes an ontology-grounded semantic entity retrieval framework for digitized Traditional Vietnamese Medicine literature, integrated with a chatbot interaction layer. The system performs semantic restructuring by converting historical documents into an ontology-compliant knowledge graph encompassing over 700 medicinal plants. Unlike systems operating on raw textual contexts, this graph explicitly models medical entities (e.g., herbs, formulas, diseases) and their therapeutic relationships. Operating on a decoupled multi-agent architecture, the system translates natural language into executable queries, enabling precise graph-based reasoning. This approach enforces an evidence-grounded generation pipeline, strictly bounding responses to the structured semantic space to substantially mitigate hallucination risks. Evaluated using the Ragas methodology, the framework demonstrates promising semantic retrieval reliability and ontology consistency. Achieving a Faithfulness score of 0.8101 and a Context Recall of 0.8251, the system empirically proves its effectiveness in executing complex, multi-hop medical retrieval tasks compared to standard conversational AI models.

**Keywords:** Semantic Retrieval; Knowledge Graph; Ontology; GraphRAG; Traditional Medicine.

## 1 Introduction

Traditional Vietnamese Medicine (TVM) is a valuable knowledge repository that plays an important role in public healthcare through the use of natural herbs and traditional remedies (World Health Organization, 2013). However, access to this knowledge remains limited, as most information is still stored in unstructured historical medical texts. Consequently, knowledge about medicinal ingredients is often fragmented and lacks explicit connections, making it difficult to retrieve complex relationships such as herbal combinations, pharmacological interactions, and therapeutic effects (Li et al., 2020). In practice, existing retrieval systems mainly rely on keyword-based search, which frequently produces incomplete results and fails to capture semantic relationships. Alternatively, generic Large Language Models (LLMs) enable more natural interactions but introduce the risk of hallucination, potentially generating unverified medical information (Ji et al., 2023). These limitations highlight the critical need for systems capable of preserving structured medical knowledge while supporting reliable semantic querying.

To address this gap, this paper proposes an ontology-grounded semantic medical knowledge retrieval framework for TVM, supported by a conversational interaction layer. Instead of operating directly on raw textual contexts, the system semantically restructures digitized historical medical texts. This approach explicitly represents dialectical relationships among herbs, formulas, active ingredients, and diseases within a strictly defined ontology. By integrating this structured semantic space with a controlled reasoning mechanism, the framework ensures more accurate, multi-hop, and traceable information retrieval. This study utilizes a dataset of over 700 Vietnamese medicinal plants extracted and standardized from authoritative medical literature.

The main contributions of this paper are fourfold. Firstly, regarding the semantic restructuring of TVM knowledge, we develop a Medallion data pipeline that transforms unstructured historical texts into an ontology-driven knowledge graph, facilitating structured medical knowledge representation. Secondly, we implement a graph-based reasoning and retrieval mechanism supporting explicit therapeutic reasoning and multi-hop queries across complex medical entity relationships. Thirdly, to ensure constrained evidence-grounded response generation, we design a generation module that strictly anchors outputs in verified graph data to substantially mitigate hallucination risks and improve factual consistency. Finally, we integrate a chatbot interaction layer specifically tailored for querying and translating structured medical knowledge into accessible formats for users. Furthermore, we evaluate the framework focusing on semantic retrieval reliability, ontology consistency, and faithfulness, with results empirically demonstrating the system's capability to handle complex medical reasoning tasks within evaluated settings, without relying on unverified generative assumptions. The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 describes the proposed methodology and system architecture, Section 4 presents the experimental results and discussion, and Section 5 concludes the paper and outlines future work.

## 2 Related Works

On a global scale, the modernization of Traditional Medicine, particularly Traditional Chinese Medicine (TCM), has made significant progress through the development of digital knowledge ecosystems (Wang et al., 2021). Recent efforts have focused on building large-scale knowledge bases such as TCMSP (Ru et al., 2014) and SymMap (Wu et al., 2019), which integrate information ranging from medicinal compounds to molecular targets. A notable trend in this area is the transition from traditional relational databases to graph-based knowledge representation (Hogan et al., 2021). Knowledge Graphs enable explicit modeling of complex relationships and support semantic reasoning and multi-hop queries, which are essential for traditional medicine applications (Yu et al., 2017). In parallel, recent studies have explored the integration of Knowledge Graphs with Large Language Models (LLMs) through Retrieval-Augmented Generation (RAG) to improve query flexibility and reduce hallucination (Lewis et al., 2020; Ji et al., 2023).

In Vietnam, the digitization of medicinal knowledge has also seen progress, with contributions from institutions such as the National Institute of Medicinal Materials (2017) and community-driven platforms like Vietherb (Nguyen & Tran, 2022). These systems play an important role in preserving indigenous knowledge by providing structured descriptions, images, and usage information (Do, 2004). However, these current platforms exhibit several technical limitations. Firstly, most systems rely heavily on keyword-based search, requiring exact input of medicinal names, which is problematic due to the presence of synonyms and linguistic variations in Vietnamese (Vu et al., 2022). Secondly, the lack of semantic connectivity between data entries prevents the system from supporting complex queries such as multi-hop reasoning. Finally, relationships among herbs, formulas, and treatments are not explicitly modeled, limiting the system’s ability to support automated reasoning and semantic relation disambiguation.

These observations reveal a clear research gap, as existing systems either lack a rigorous semantic structure or rely on generic generative models that are susceptible to unverified outputs. Therefore, there remains a critical need for a comprehensive framework that combines ontology-driven knowledge representation with constrained semantic retrieval for Traditional Vietnamese Medicine. To address this gap, this study proposes an ontology-grounded semantic medical knowledge retrieval framework, supported by a chatbot interaction layer. Rather than simply applying natural language processing for text-to-query translation, the framework systematically restructures unstructured historical records into a highly interconnected, ontology-compliant Knowledge Graph comprising over 700 digitized medicinal plants (Do, 2004). By integrating this standardized semantic space with graph-based reasoning algorithms, the system ensures that retrieved information is strictly evidence-grounded, supporting accurate multi-hop retrieval while substantially reducing hallucination risks within evaluated settings.

### 3 Proposed Methodology

#### 3.1 3-Agent GraphRAG Architecture

Unlike conventional text-based Retrieval-Augmented Generation (RAG) systems or generic conversational agents that bundle natural language processing and generation into a single Large Language Model (LLM), this study proposes an ontology-grounded, functionally decoupled multi-agent architecture. This framework is specifically designed to enforce semantic control and substantially mitigate hallucination risks. The system operates on a three-agent workflow strictly integrated with the semantic knowledge graph, functioning as a structured graph-based reasoning framework engine rather than a mere conversational tool.

First, the User Query is ingested by Agent 1: NLU Parser, which performs semantic intent analysis and extracts key medical entities mapped to the ontology. The parsed request is then routed to the Retrieval Module, where the Cypher Builder acts as a central orchestrator. It simultaneously triggers distinct search mechanisms, including Exact Match, Semantic Matching, and Constraints Search, directly against the Knowledge Graph database. Once the graph reasoning process is executed, the retrieved raw results pass through a Result Filter. This step aims to significantly reduce background noise and standardize the retrieved entities into a highly structured therapeutic representation.

Finally, the refined semantic data enters the Generation and Safety cluster, establishing a dual-control phase crucial for system reliability. Here, Agent 2: Generator synthesizes an initial draft response grounded strictly in the retrieved graph evidence. Immediately afterward, this draft is cross-validated by Agent 3: Evaluator. By verifying generated claims against the ground-truth facts, the evaluating agent detects and filters out unverified information. The system outputs the final response only if it is closely aligned with the source data. Ultimately, this decoupled architecture ensures that generated responses are backed by verifiable evidence, empirically improving the factual consistency of the medical knowledge retrieval framework.

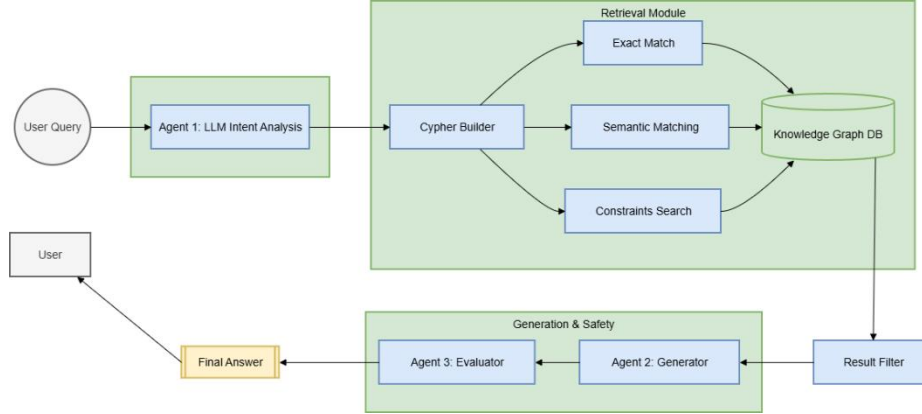


Figure 1. 3-Agent GraphRAG Architecture.

To address the unstructured, polysemous, and inconsistent nature of historical Traditional Medicine data, this research implements a multi-layered semantic restructuring pipeline based on the Medallion architecture. The data is sequentially refined through three primary stages—Bronze (raw ingestion), Silver (semantic extraction), and Diamond (ontology alignment)—empirically improving data integrity and semantic retrievability.

The Bronze layer focuses on converting physical historical records into digitized text. The foundational dataset of 712 medicinal plants was extracted from the authoritative literature, *Medicinal Plants and Remedies in Vietnam* (Do, 2004). To mitigate context loss during text chunking, the pipeline segments data using semantic anchors and regular expressions, precisely determining the boundaries of individual herbal entries. Each extracted segment is stored alongside source metadata, including page numbers and document origins, to support evidence-grounded traceability during the generation phase.

In the Silver layer, unstructured text is transformed into a structured format using Large Language Models (LLMs) constrained by predefined ontology schemas. To optimize this process, the framework applies a "divide-and-conquer" strategy, modularizing the extraction into independent semantic groups: identification details, formulas, active ingredients, and pharmacology. This decomposition mitigates the impact of LLM context window constraints, promotes strict adherence to the defined data schema, and enables isolated debugging of specific information clusters. Consequently, semantic extraction deviations can be systematically detected and corrected, substantially reducing structural errors.

The Diamond layer serves as the final semantic normalization and validation phase before ingestion into the graph database. This stage employs deterministic rule-based algorithms to improve relation disambiguation and ontology consistency. Specifically, the framework normalizes entity identifiers, constructs a master synonym dictionary to resolve nomenclature variations, and audits relational links to identify orphan nodes or invalid connections. Furthermore, quantitative metrics, such as the Data Richness Index (DRI), evaluate the semantic completeness of each entity. The output

is a rigorously structured, ontology-compliant dataset optimized for explicit graph-based reasoning.



Figure 2. Professional Data Pipeline based on the Medallion Architecture.

### 3.2 Ontology Design and Knowledge-Enriched Embedding

Traditional Medicine (TM) knowledge is highly dialectical, characterized by complex, multi-layered relationships between entities such as herbs, formulas, symptoms, and mechanisms of action. Conventional Relational Database Management Systems (RDBMS) face significant constraints when handling multi-hop queries or requiring flexible schema expansions for unstructured medical texts (Noy & McGuinness, 2001). To address this, an ontology-driven knowledge representation offers fundamental advantages.

First, ontology enables the modeling of entities and relationships that align with the philosophical structure of TM, such as the principles of "Monarch-Minister-Assistant-Guide." This facilitates semantic normalization, allowing the framework to represent medical logic contextually rather than relying on superficial keyword matching (Yu et al., 2017).

Second, within a Knowledge Graph framework, relationships are explicitly structured and directly queryable. This inherently supports multi-hop reasoning tasks—such as identifying herbs with identical properties but different meridian

tropisms—which are computationally expensive in traditional SQL environments (Hogan et al., 2021). Third, the ontology functions as a rigorous semantic constraint layer, ensuring retrieved information is anchored in validated relationships. By providing a structured therapeutic representation, this approach substantially reduces hallucination risks in systems integrated with Large Language Models (LLMs), prioritizing factual consistency over generative assumptions (Pan et al., 2024).

Based on the digitized dataset of 712 medicinal plants, a conceptual schema was developed encompassing nine primary entity nodes linked by directed edges. These entities are systematically categorized into three semantic layers: core entities (Herb, Formula, Disease); traditional medicine attributes (Symptom, Property, Flavor, Meridian, Function); and modern pharmacological attributes (Active Ingredient, Pharmacology). The relationships among these nodes simulate the dialectical logic of TM prescriptions. This strict schema enforces relation disambiguation, ensuring each structural link carries a precise semantic weight necessary for explicit graph-based reasoning, as detailed in Table 1.

Table 1. Relationship matrix in the TVM Knowledge Graph.

| Relationship Name   | Source         | Target            | Semantic Meaning                                               |
|---------------------|----------------|-------------------|----------------------------------------------------------------|
| BAO_GOM_VI_THUOC    | Formula        | Herb              | Represents ingredients and compatibility structure             |
| CHU_TRI_BENH        | Herb / Formula | Disease           | Therapeutic indications for reverse querying from diseases     |
| CHU_TRI_TRIEU_CHUNG | Herb / Formula | Symptom           | Therapeutic indications specifically for alleviating symptoms. |
| CO_TINH             | Herb           | Property          | Thermal characteristics (Cold, Hot, Warm, Cool)                |
| CO_VI               | Herb           | Flavor            | Taste profiles (Pungent, Bitter, Sweet, etc.)                  |
| QUY_KINH            | Herb           | Meridian          | Action pathways toward internal organs                         |
| CO_CONG_NANG        | Herb / Formula | Function          | Effects based on TM theoretical reasoning                      |
| CO_CHUA_HOAT_CHAT   | Herb           | Ingredient        | Chemical composition                                           |
| CO_TAC_DUNG_DUOC_LY | Herb           | Pharmacology      | Biological effects according to modern medicine                |
| KIENG_KY            | Herb / Formula | Patient / Disease | Contraindications and safety warnings.                         |

To mitigate ambiguity between entities with similar names or characteristics, the framework advances beyond isolated keyword-based representations. Instead, each entity is represented through an expanded semantic context. Specifically, the text used for entity representation is a synthesized string comprising the entity's canonical name, aliases, a curated search hint, and its structural relationships with neighboring nodes within the ontology.

This comprehensive descriptive string is then transformed into a semantically rich vector space, allowing for similarity searches based on contextual patterns rather than simple character sequences. When integrated with the framework's retrieval mechanism, this approach substantially improves entity identification across synonymous or non-standardized nomenclature, ensuring that retrieved nodes are strictly aligned with the defined medical ontology (Sun et al., 2019).

To interact with the ontology-driven knowledge graph, the framework implements a Semantic Parsing and Graph-Based Reasoning layer. Rather than treating user inputs as simple text retrieval triggers, the workflow begins at the Semantic Analysis

module. This module utilizes few-shot prompting techniques to map natural language into precise semantic constraints, extracting three core components: the intent, the primary entity, and the constraint entity. To optimize computational efficiency, a local caching mechanism stores recurring query results. The transformation from natural language to executable graph queries then proceeds through a structured sequence.

Initially, text undergoes NFKD normalization to remove diacritics and synchronize formatting (e.g., “Ích mẫu” to ICH\_MAU), facilitating consistent node alignment within Neo4j. Subsequently, the framework utilizes Cypher templates to map the normalized input based on the identified intent. A schema mapping table defines relationship types and link directions, categorizing queries into fundamental attribute retrieval, multi-hop reasoning, and Boolean logic verification. For complex queries involving multi-condition constraints, filtering conditions are pushed directly into the Cypher WHERE clause. This filter push-down approach allows logical intersections to be executed efficiently at the database level (Francis et al., 2023). Finally, the system executes a hybrid search combining exact string matching with contextual semantic matching. Merging these results via the UNION operator enhances both retrieval coverage and precision (Gao et al., 2023).

Table 2. Representative Intent groups and Cypher mapping logic.

| Intent Group        | Example Intent Code | Cypher Macro Logic                    | Inference Goal                                 |
|---------------------|---------------------|---------------------------------------|------------------------------------------------|
| Entity Retrieval    | TIM_VI, TIM_TINH    | MATCH (s)-[r:CO_VI/TINH]->(t)         | Extracting original herbal properties          |
| Formula Analysis    | TIM_THANH_PHAN      | (BT)-[:BAO_GOM_VI_THUOC]->(VT)        | Decoding formula composition                   |
| Reverse Inference   | TIM_THUOC_CHUA_BENH | (VT/BT)-[:CHU_TRI_BENH]->(B)          | Finding treatments based on symptoms           |
| Logic Verification  | KIEM_TRA_BOOLEAN    | MATCH (s)-[r]->(t) WHERE t.id = \$k2  | Validating compatibility and interactions      |
| Multi-layered Query | TIM_DA_QUAN_HE      | WHERE s.tinh = \$k1 AND t.benh = \$k2 | Executing logic filters on multiple conditions |

Upon retrieving results from the Neo4j knowledge graph, the framework executes a post-processing phase to standardize the output prior to natural language generation. Initially, during the Fact-Sheet construction phase, the retrieved JSON data is transformed into a structured set of concise propositions. This step preserves critical quantitative details, including dosage, usage instructions, and pharmacological characteristics. Consequently, this Fact-Sheet functions as the explicit grounding reference—serving as the "sole source of truth"—for the subsequent generation stage (Lewis et al., 2020).

Following data standardization, an evidence-grounded generation process is executed under rigorous parameters. Specifically, the language model's temperature is configured to 0.0 to promote deterministic outputs, thereby reducing generative variability. The synthesized content is strictly bounded by the scope of the Fact-Sheet, mitigating the risk of incorporating unverified external knowledge. Furthermore, the conversational interaction layer integrates a fallback mechanism; if exact data is unavailable, it retrieves closely related entities rather than generating speculative information. This design constraint supports the framework's operational objective as a highly regulated semantic medical retrieval system.

## 4 Experiments

### 4.1 Experimental Setup

To meet the stringent safety requirements of the healthcare domain, the framework is implemented based on the proposed decoupled 3-Agent GraphRAG Architecture. Operating on a "Safety-First" principle, the system comprises three distinct processing clusters which isolate specific tasks to restrict the system's reliance on the free-form reasoning capabilities of Large Language Models (LLMs), thereby systematically improving the factual integrity of the retrieved medical literature. These clusters include a semantic parser dedicated to intent recognition and entity extraction, a Neo4j knowledge graph utilizing structured graph traversal as a factual anchor, and a generation module equipped with strict semantic filtering. To evaluate this ontology-grounded architecture across different processing layers, the empirical validation is formally divided into two distinct core experiments.

The first experiment focuses on validating the semantic extraction efficacy of the proposed Medallion data pipeline during the Silver layer processing. The foundational experimental data for this phase consists of digitized medical texts from the Bronze layer, which were randomly sampled from the primary corpus constructed from 712 traditional medicinal plant species. This setup employs a rigorous A/B testing methodology to benchmark two distinct approaches, directly comparing the proposed modular divide-and-conquer pipeline against a Single-pass Baseline that prompts the LLM to extract all entities and relationships simultaneously. The resulting extraction quality is quantitatively evaluated through an LLM-as-a-Judge approach, which measures the structured outputs against the raw OCR ground truth based on three core technical metrics including Schema Compliance, Total Entity Recall, and Hallucination Rate.

The second experiment is designed to comprehensively evaluate the framework's end-to-end semantic retrieval capabilities and evidence-grounded generation performance. Evaluating this ontology-grounded architecture requires a tailored approach because a direct comparison with generic conversational AI or standard text-based RAG baselines may not fully reflect the system's performance, given the fundamental differences in semantic representation spaces and the strict prioritization of constrained, factual adherence over conversational fluency. To execute this benchmark, the evaluation framework utilizes a specialized test dataset comprising 240 automatically generated queries that span complex user intent scenarios such as attribute lookups, multi-hop pathological reasoning, Boolean logic verification, and multi-constraint intersection queries. Rather than relying on superficial exact-string matching metrics such as traditional Precision and Recall, this end-to-end pipeline adopts the advanced Ragas (Retrieval Augmented Generation Assessment) methodology powered by the Gemini 2.0 Flash model acting as an independent judge across five core dimensions, specifically Faithfulness, Context Recall, Context Precision, Answer Correctness, and Answer Relevancy.

Regarding the technical deployment and environment setup, all data ingested into the graph underwent a rigorous preprocessing pipeline for identity normalization and

entity linking to systematically mitigate academic vocabulary ambiguities. The centralized knowledge graph was deployed using Neo4j Community Edition version 5.1, while the natural language generation module utilized an instruction-tuned LLM configured with a strict temperature of 0.0 to ensure deterministic outputs and suppress generative variability. All structural graph traversals, semantic parsing workflows, and data processing pipelines were executed locally on a dedicated hardware workstation equipped with an NVIDIA GeForce RTX 3060 GPU and 16 GB of RAM to handle the multi-hop reasoning tasks efficiently.

#### 4.2 Knowledge Extraction Performance

To rigorously validate the effectiveness of the modular divide-and-conquer strategy at the Silver layer, an A/B testing experiment was conducted on 10 randomly selected medicinal herb samples. The evaluation utilized an advanced LLM-as-a-Judge automated framework powered by the Gemini 2.0 Flash model, executing a forensic medical data audit. This automated judge was constrained by a strict prompt evaluating five dimensions: comprehensive entity coverage to penalize "lazy generation", directional ontology integrity to prevent relational reversals (e.g., erroneously predicting herbs comprising formulas), forensic traceability through exact textual claims, technical formatting discipline for scientific units, and strict penalties for structural hallucinations such as fabricated dosages. The generated structured JSON files from both the Baseline and the proposed Pipeline were programmatically evaluated against the raw OCR ground truth text.

To establish a mathematically rigorous evaluation for this knowledge extraction phase, three formal metrics were defined. Firstly, the JSON Schema Compliance quantifies the strict adherence of the generated structural outputs to the predefined ontology templates, calculated as the ratio of successfully parsed JSON structures ( $N_{\text{compliant}}$ ) over the total number of processed test samples ( $N_{\text{total}}$ ) expressed as a percentage:

$$\text{JSON Schema Compliance} = \frac{N_{\text{compliant}}}{N_{\text{total}}} \times 100 \ \%$$

Secondly, the Total Entity Recall measures the comprehensive information coverage by calculating the intersection of correctly extracted medical entities ( $E_{\text{extracted}}$ ) against the manual expert-verified ground truth ( $E_{\text{ground\_truth}}$ ):

$$\text{Total Entity Recall} = \frac{|E_{\text{extracted}} \cap E_{\text{ground\_truth}}|}{|E_{\text{ground\_truth}}|}$$

Thirdly, the Hallucination Rate is introduced to monitor structural reliability by calculating the proportion of fabricated entities, unverified dosages, or logically reversed relationships ( $E_{\text{hallucinated}}$ ) over the total generated output volume:

$$\text{Hallucination Rate} = \frac{E_{\text{hallucinated}}}{E_{\text{extracted}}} \times 100 \ \%$$

The aggregate evaluation metrics computed by the automated auditor are summarized in Table 3.

Table 3. Aggregate performance of knowledge extraction over 10 test samples

| Evaluation Metric            | Baseline (Single-pass) | Proposed (Pipeline) | Improvement  |
|------------------------------|------------------------|---------------------|--------------|
| Successfully Processed Files | 10/10                  | 10/10               | -            |
| JSON Schema Compliance       | 100%                   | 100%                | -            |
| Total Entities Extracted     | 151                    | 313                 | +107.28%     |
| Total Hallucination Errors   | 3                      | 0                   | Reduced to 0 |

**Discussion.** The empirical results demonstrate that the modular Medallion architecture provides a substantial improvement over the traditional Single-pass prompting method. In terms of knowledge coverage, the proposed pipeline extracted 313 entities, representing a 107.28% increase compared to the Baseline’s 151 entities. This significant deficiency in the Baseline illustrates the phenomenon of "Lazy Generation"—a critical cognitive bottleneck where large language models suffer from context overload when forced to simultaneously manage extensive, unstructured medical texts and strict JSON serialization schemas. By systematically decomposing the extraction task into specialized, independent modules focusing distinctly on identification, formulas, active ingredients, and pharmacology, the proposed framework allows the underlying model to maintain granular focus, substantially boosting the recall of intricate chemical constituents.

Furthermore, the architectural design establishes absolute reliability and hallucination control. While the Baseline recorded critical logical hallucinations—such as reversing structural entity relationships and generating unverified contraindications—the proposed pipeline achieved a zero hallucination rate within the evaluated samples. This stability is driven by the strict ontology guidelines that systematically eliminate semantic drift. Crucially, this pipeline delivers complete medical traceability through integrated source metadata and claim validation arrays. This structural traceability allows every retrieved node and edge inside the final Neo4j database to be forensically traced back to its precise historical OCR origin, effectively fulfilling the stringent transparency mandates required for evidence-grounded clinical knowledge retrieval frameworks.

### 4.3 Semantic Retrieval and Evidence-Grounded Generation Performance

The evaluation results of the framework on the 240-query test set, utilizing the Ragas methodology, are detailed in Table 4. Notably, this study adopts an absolute benchmarking approach rather than a comparative evaluation against a standard text-based RAG baseline. The rationale for this methodological choice stems from the structural incompatibility of conventional vector-similarity search with the designed test suite. Because the 240 queries heavily feature multi-hop pathological reasoning and global aggregations (e.g., retrieving all herbs sharing a specific property scattered across the entire corpus), a standard RAG architecture is inherently incapable of compiling the necessary context due to the severe information fragmentation caused by Top-K chunking. Consequently, evaluating a standard baseline on these complex structural queries would yield near-zero recall, rendering a direct numerical comparison scientifically asymmetric and uninformative. Thus, the empirical scores in Table 4 serve to quantitatively validate the standalone efficacy of the proposed ontology-grounded architecture.

Table 4. Performance evaluation of the GraphRAG system using Ragas metrics (N=240)

| Ragas Metrics      | Average Score |
|--------------------|---------------|
| Context Recall     | 0.8251        |
| Faithfulness       | 0.8101        |
| Context Precision  | 0.7941        |
| Answer Correctness | 0.5992        |
| Answer Relevancy   | 0.4628        |

**Discussion.** Regarding semantic retrieval performance, the framework achieved Context Recall and Context Precision scores of 0.8251 and 0.7941, respectively. These results empirically validate the efficacy of the structured graph traversal mechanism over traditional text retrieval approaches. By dynamically aggregating fragmented medical relationships into high-density 'Fact-Sheets' prior to generation, the system successfully extracts critical multi-hop information while substantially reducing background noise that could otherwise impede the semantic reasoning process.

In terms of hallucination control, Faithfulness (0.8101) serves as a paramount metric for evaluating the factual consistency of medical knowledge frameworks. Exceeding the 0.8 threshold indicates that the ontology-grounded multi-agent architecture operates with substantial stability. The integration of the evaluating agent, alongside a zero-temperature configuration, functions as a rigorous semantic guardrail. This setup strictly constrains the generation module to align with the deterministic data extracted from Neo4j, thereby significantly reducing hallucinated responses before they reach the user interface.

Regarding generation quality, Answer Correctness maintained a moderate level (0.5992), and Answer Relevancy scored 0.4628. These lower relevancy scores are not indicative of system failure, but rather reflect a deliberate architectural trade-off inherent to evidence-grounded generation. Generic LLMs typically generate verbose narrative fillers that artificially inflate Relevancy metrics in standard evaluations. Within this highly regulated healthcare framework, prioritizing strict factual adherence and semantic normalization over conversational verbosity is a rational necessity to support reliable medical retrieval.

Finally, a significant architectural constraint involves the processing of the 'KIENG\_KY' (Contraindication) relationship. Although generic generative models possess the capacity to infer potential contraindications from general herbal properties, AI-driven reasoning was deliberately suppressed for this sensitive domain. This constraint restricts the framework to providing only those contraindication warnings explicitly documented in the authoritative medical literature. Prioritizing structural evidence over generative assumptions substantially mitigates the risk of severe clinical hallucinations, acknowledging the trade-off of potentially sparser responses for undocumented contraindication queries.

## 5 Conclusion

This study proposes an ontology-grounded semantic medical knowledge retrieval framework for Traditional Vietnamese Medicine, supported by a decoupled 3-Agent architecture. By separating semantic parsing and evidence-grounded generation, the

framework utilizes an ontology-driven Knowledge Graph as the primary source of factual information, thereby substantially mitigating hallucination risks commonly observed in generic Large Language Models. The system integrates a multi-stage Medallion data restructuring pipeline, a rigorous ontology to map complex relationships, and a controlled response generation process based on a Fact-Sheet mechanism. Overall, the empirical results demonstrate that combining LLMs with structured Knowledge Graphs improves semantic consistency and evidence-grounded medical knowledge access.

Despite these promising results, several limitations remain. The foundational Knowledge Graph is derived exclusively from digitized historical documents and has not yet undergone comprehensive verification by modern clinical experts. Additionally, the current semantic parsing module depends on deterministic template-based mapping, restricting its flexibility when processing highly ambiguous queries. Furthermore, the dataset's coverage of specific contraindications remains limited by the explicit content of the historical texts. To prioritize medical safety, the framework explicitly restricts generative AI reasoning for undocumented contraindication data. Finally, the system has not yet been evaluated in active clinical environments.

Future research will address these limitations across several dimensions. First, a 'Clinical Expert-in-the-Loop' module will be integrated, enabling TVM practitioners to manually validate, expand, and update the knowledge base without compromising graph reliability. Second, the semantic parsing module will be advanced toward neuro-symbolic approaches, allowing for dynamic ontology alignment to better handle historical ambiguities. Third, retrieval performance will be enhanced by exploring more sophisticated structured graph traversal strategies. Finally, the framework will be deployed in real-world scenarios to evaluate its practical utility and further refine its multi-hop therapeutic reasoning capabilities.

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